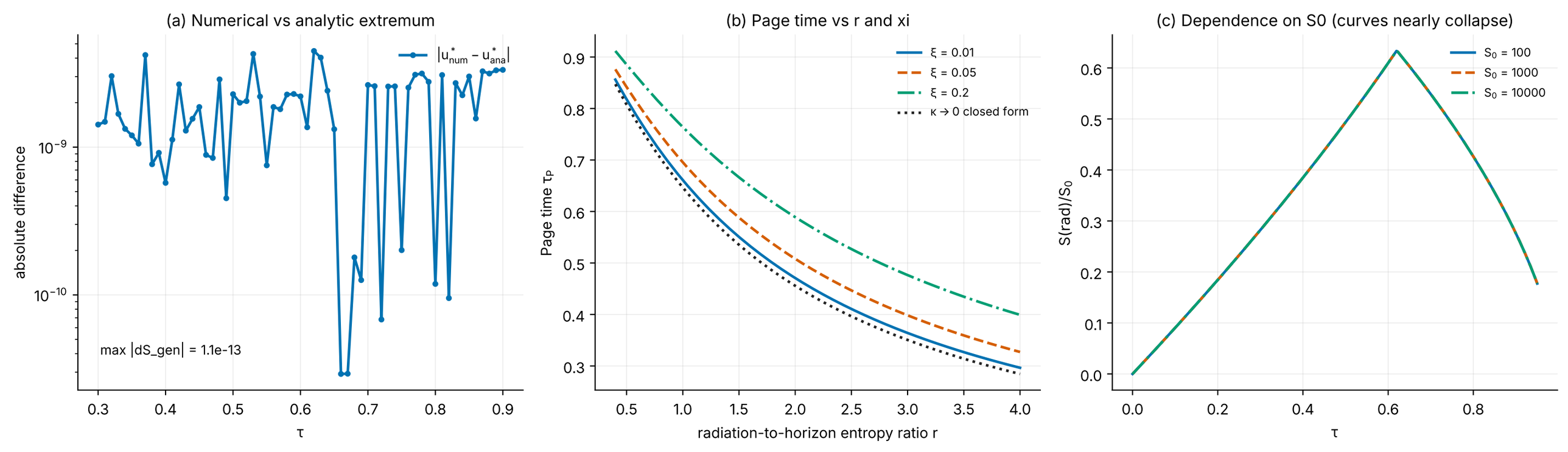
(a) Numerical vs analytic extremum
absolute difference
10^{-9}
10^{-10}
$|u^*_{num} - u^*_{ana}|$
max |dS_gen| = 1.1e-13
0.3
0.4
0.5
0.6
0.7
0.8
0.9
τ
(b) Page time vs r and xi
Page time τ_P
ξ = 0.01
ξ = 0.05
ξ = 0.2
κ → 0 closed form
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
radiation-to-horizon entropy ratio r
(c) Dependence on S0 (curves nearly collapse)
S(rad)/S_0
S_0 = 100
S_0 = 1000
S_0 = 10000
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.0
0.2
0.4
0.6
0.8
τ